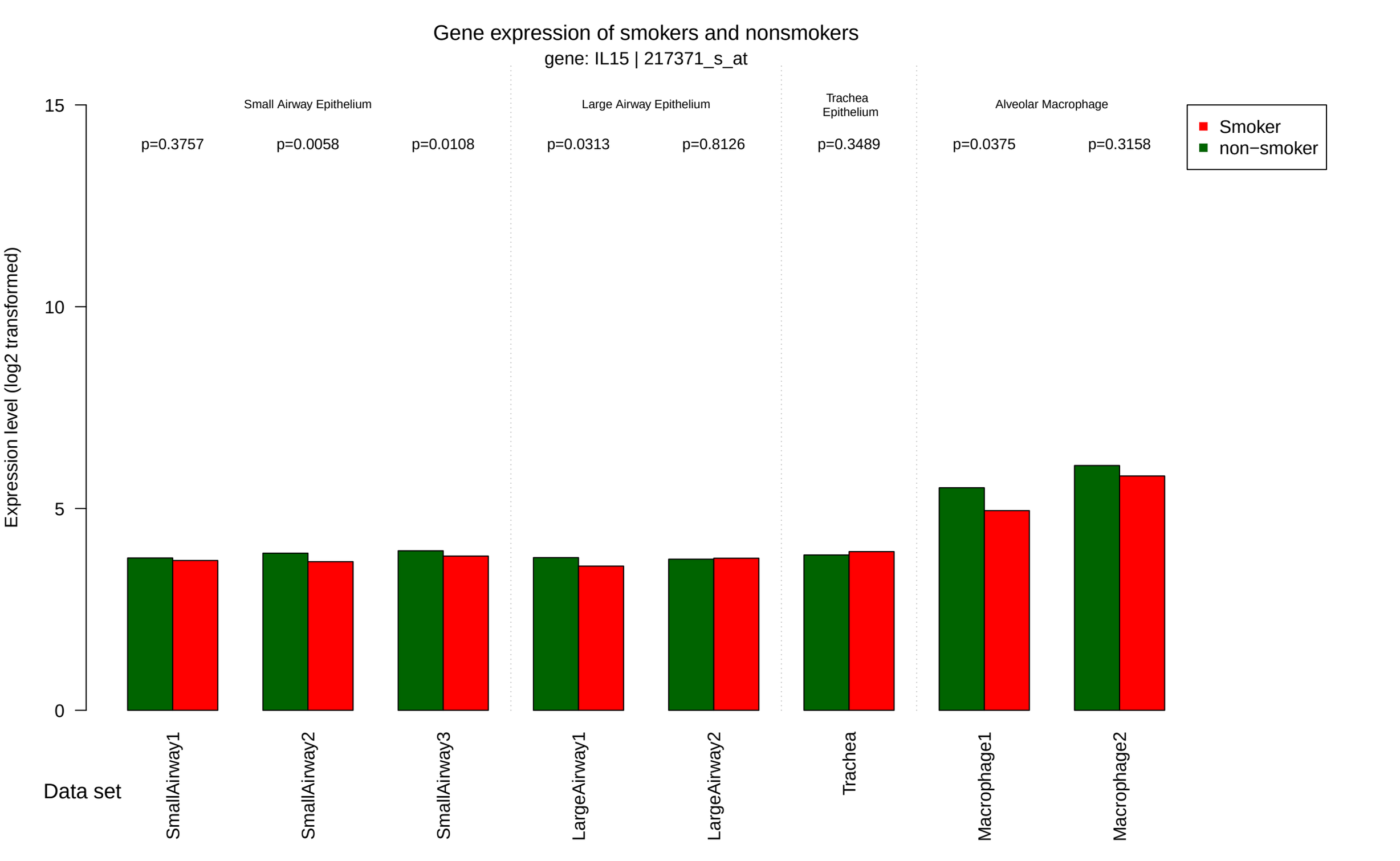
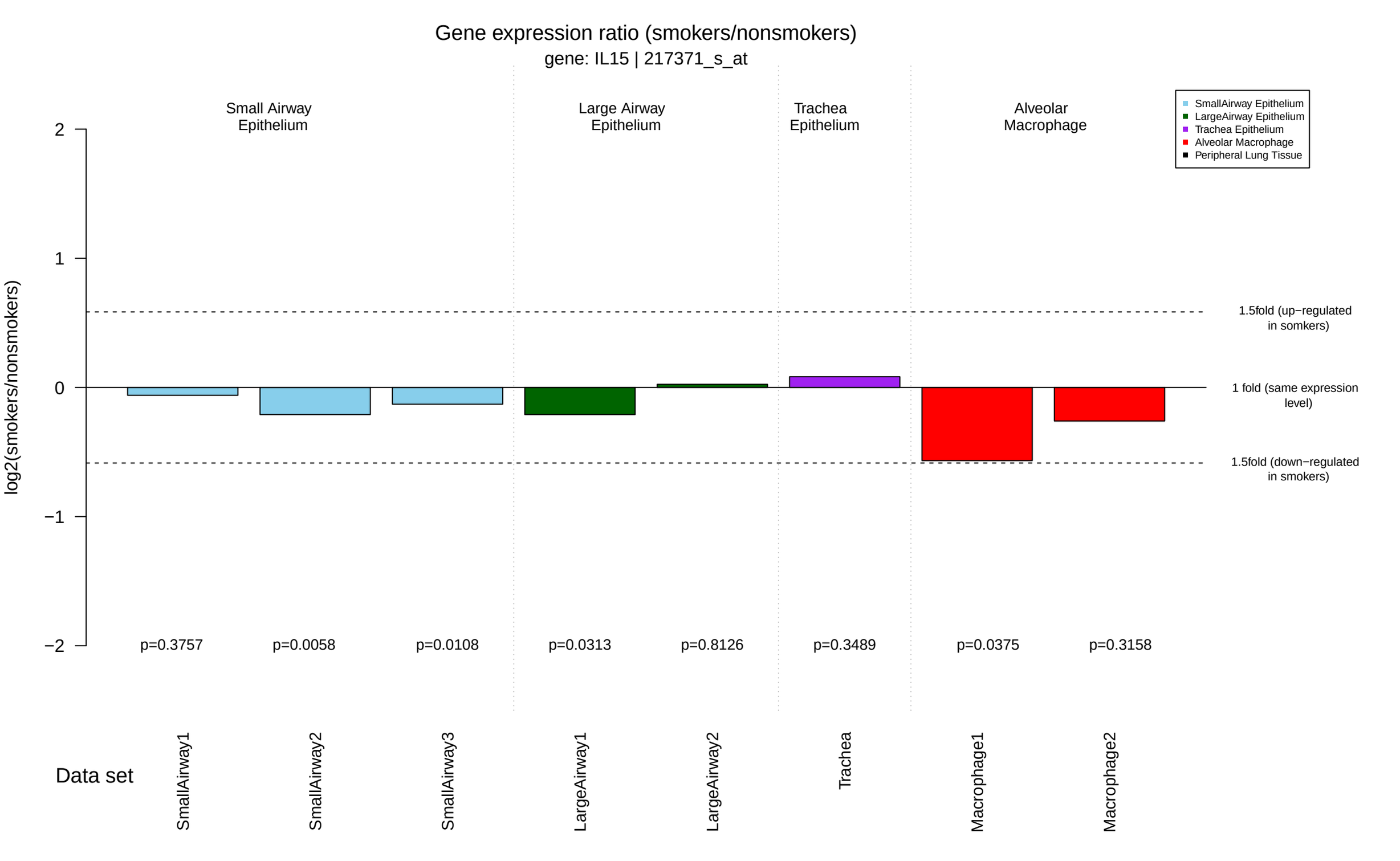
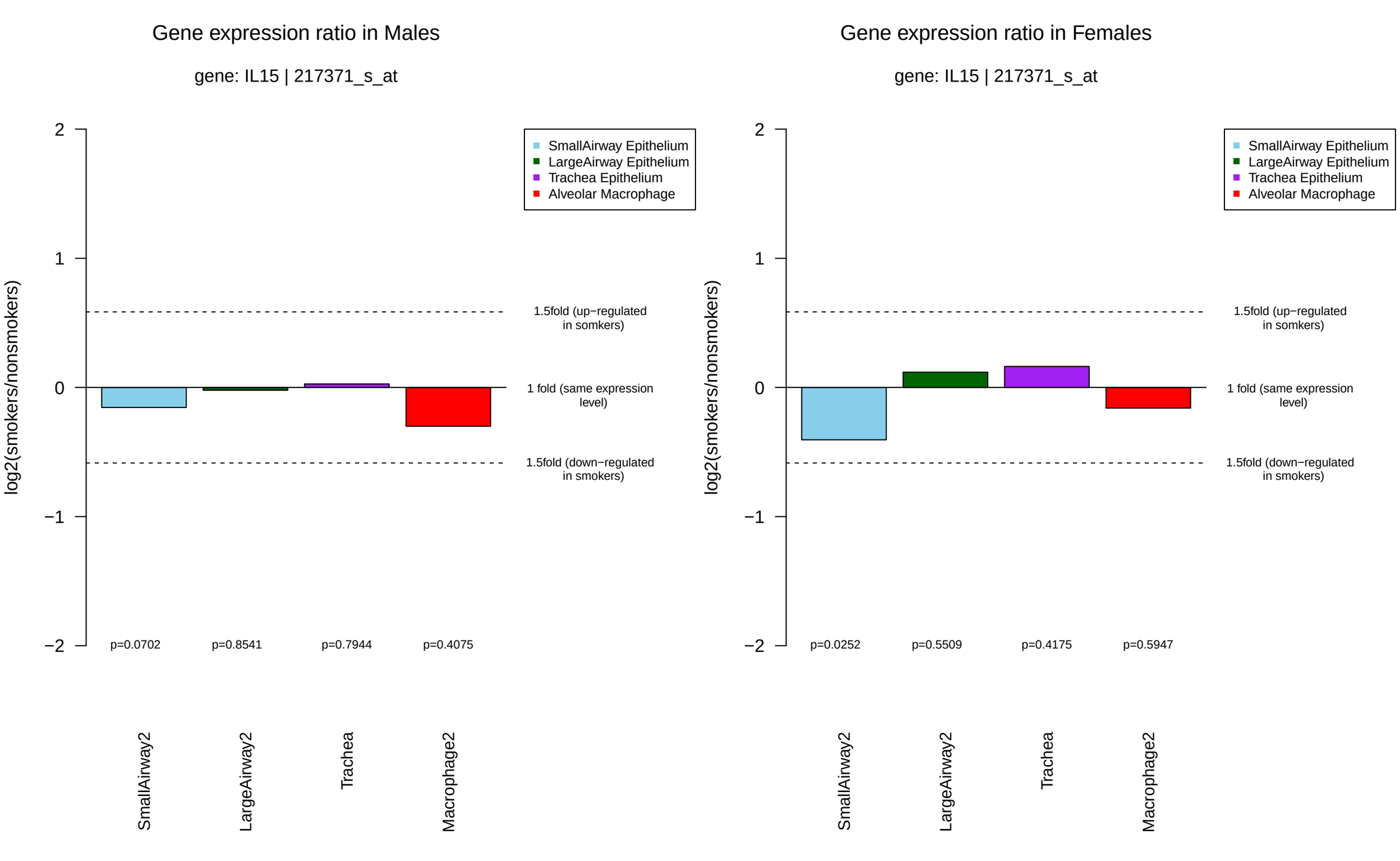
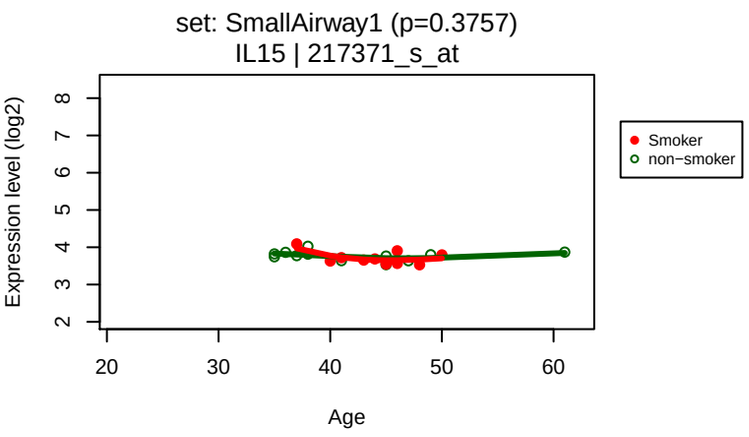
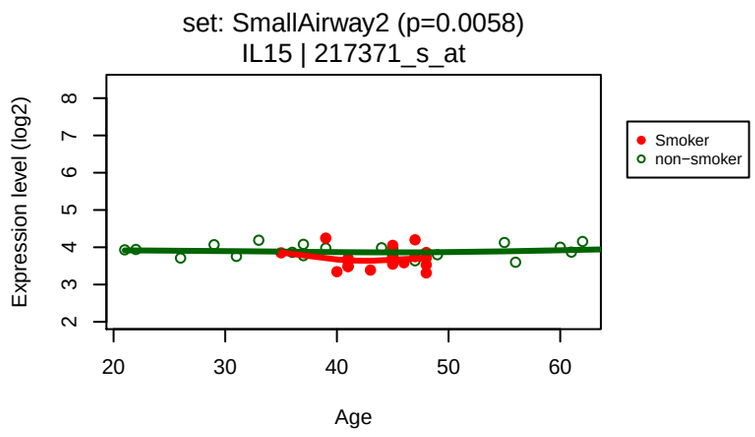
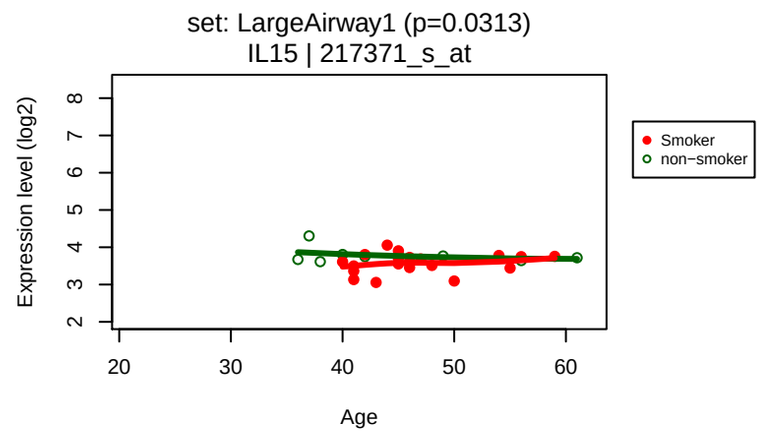

Gene expression of smokers and nonsmokers
gene: IL15 | 217371_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3757
p=0.0058
p=0.0108
p=0.0313
p=0.8126
p=0.3489
p=0.0375
p=0.3158
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

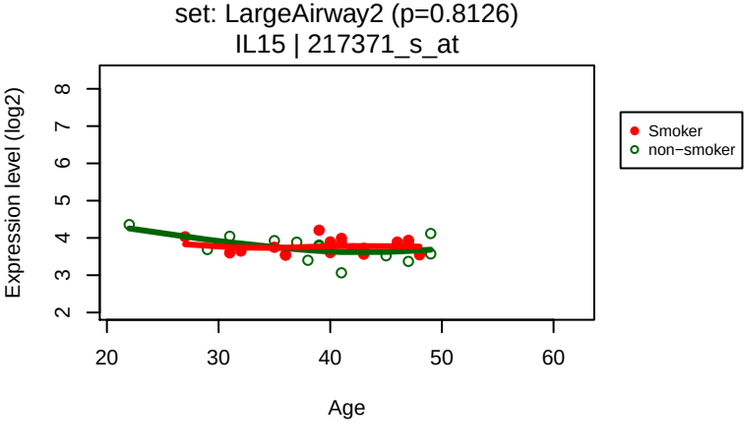

Gene expression ratio (smokers/nonsmokers)
gene: IL15 | 217371_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3757
p=0.0058
p=0.0108
p=0.0313
p=0.8126
p=0.3489
p=0.0375
p=0.3158
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

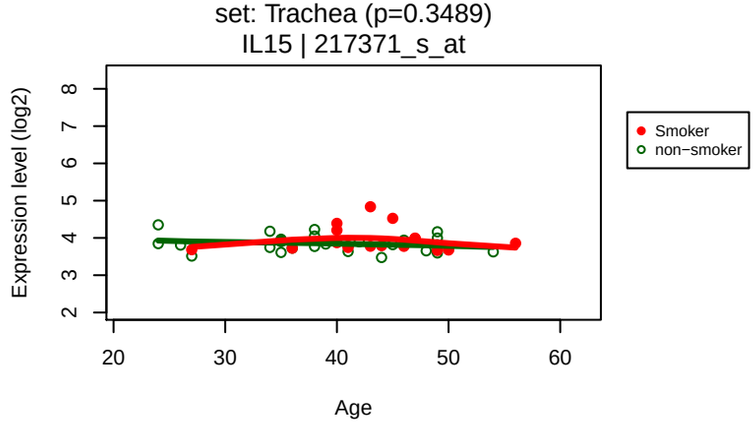

Gene expression ratio in Males
gene: IL15 | 217371_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0702
p=0.8541
p=0.7944
p=0.4075
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL15 | 217371_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0252
p=0.5509
p=0.4175
p=0.5947
SmallAirway2
LargeAirway2
Trachea
Macrophage2

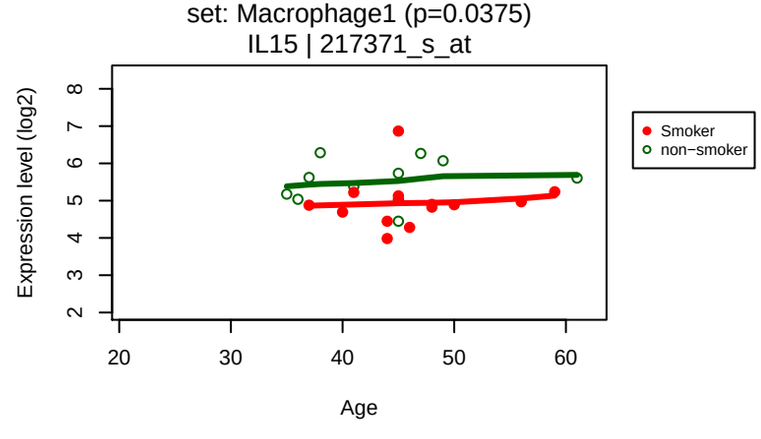

set: SmallAirway1 (p=0.3757)
IL15 | 217371_s_at
Expression level (log2)
Age
Smoker
non−smoker

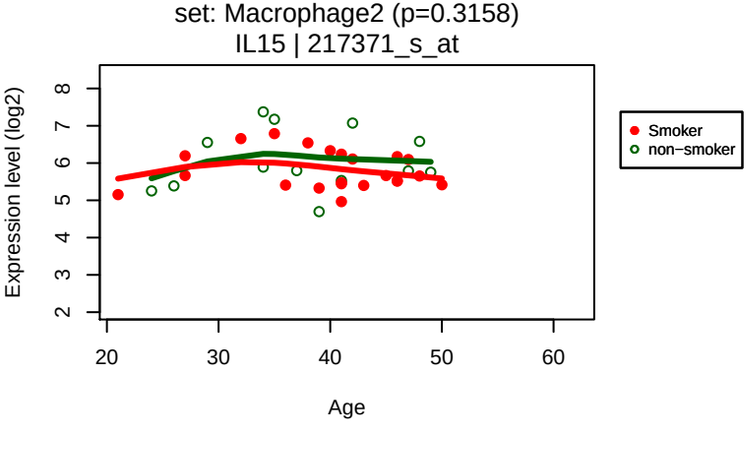

set: SmallAirway2 (p=0.0058)
IL15 | 217371_s_at
Expression level (log2)
Age
Smoker
non−smoker

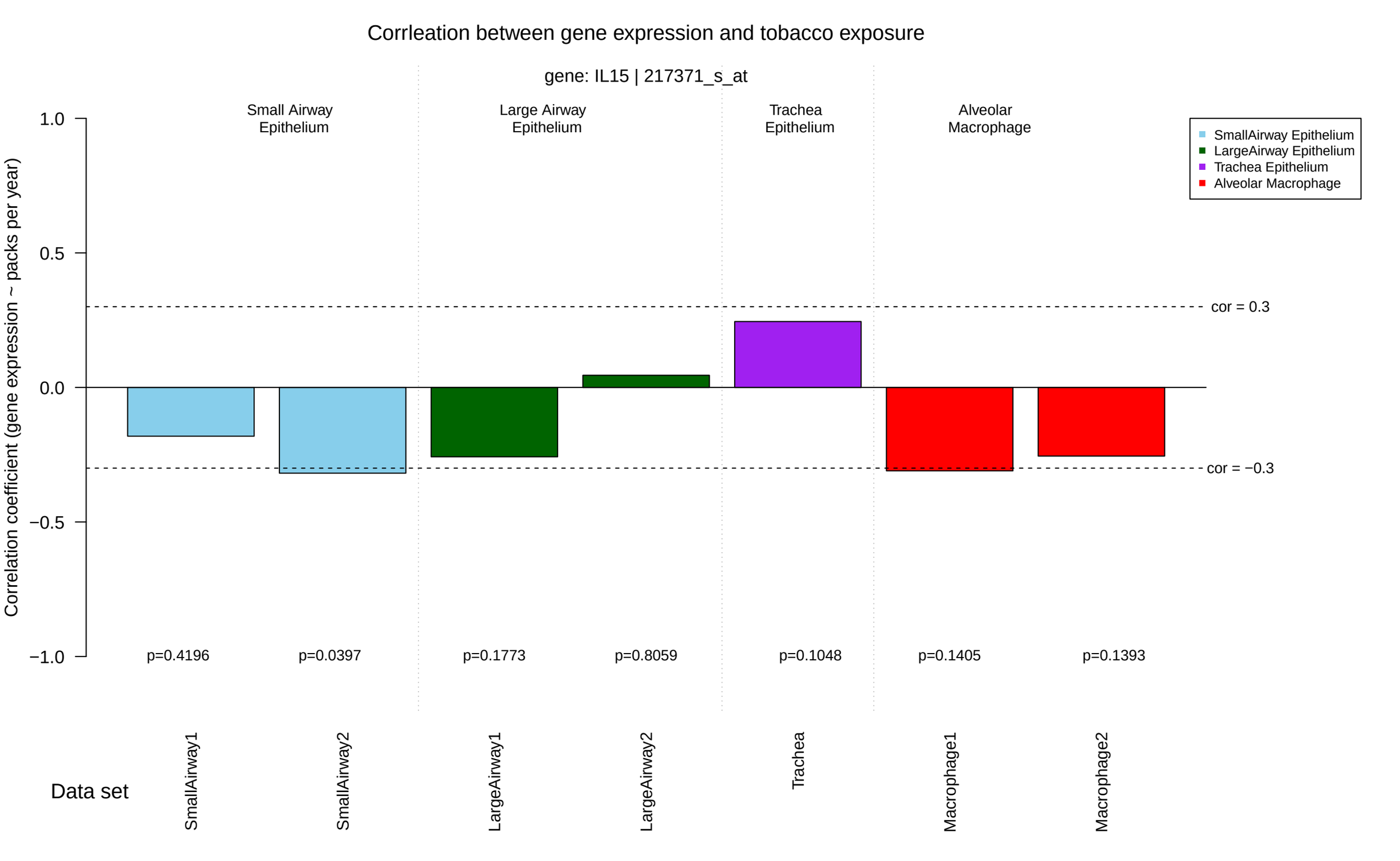

set: LargeAirway1 (p=0.0313)
IL15 | 217371_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8126)
IL15 | 217371_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3489)
IL15 | 217371_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0375)
IL15 | 217371_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3158)
IL15 | 217371_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL15 | 217371_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4196
p=0.0397
p=0.1773
p=0.8059
p=0.1048
p=0.1405
p=0.1393
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2